

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I __Shaik. Abdul Razak_(192524358)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _ Shaik. Abdul Razak _____

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

1. Cloud Architecture

The Smart Retail Inventory Management System uses a **cloud-based architecture** in which POS systems and dashboards act as clients. These clients communicate with centralized cloud services through APIs. Cloud storage maintains product information, transaction records, and analytics data.

2. Main Components

- **Clients:** POS systems and management dashboards.
- **API Gateway:** Receives and manages requests from clients.
- **Cloud Services:** Process inventory and transaction operations.
- **Cloud Storage:** Stores product data, transaction logs, and analytics datasets.
- **Platform Services:** Provide analytics and demand forecasting.
- **Security Infrastructure:** VPNs, firewalls, authentication, and API security.

3. Working

1. A customer purchases a product through the POS system.
2. The POS sends the transaction information through an API.
3. Cloud services process the transaction.
4. Inventory information is updated in cloud storage.
5. Analytics services analyze sales and inventory data.
6. Demand forecasting predicts future product requirements.
7. Dashboards display updated inventory and analytical information to managers.

4. Security

Security is maintained using:

- VPN for secure network communication.
- Firewalls to prevent unauthorized network access.
- API gateways to control API requests.
- Authentication and authorization for users and devices.
- Secure cloud storage for business data.

5. Benefits

- Real-time inventory monitoring.
- Centralized data management.
- Scalable cloud infrastructure.
- Better demand forecasting.
- Reduced manual inventory management.
- Secure access to business information.

Case Study 2: Cloud-Based Document Collaboration System

1. Cloud Architecture

The Cloud-Based Document Collaboration System provides an online platform similar to Google Docs. Users access documents through web browsers, while cloud services manage document editing, storage, synchronization, and sharing.

2. Main Components

- **Web Browsers:** Used by users to access documents.
- **Web Application:** Provides the document editing interface.
- **Web APIs:** Synchronize data between clients and cloud services.
- **Cloud Storage:** Stores documents and related data.
- **Version Control:** Maintains different versions of documents.
- **Network Infrastructure:** Provides high availability and low-latency access.
- **Security Services:** Encryption, identity management, and access control.

3. Working

1. A user logs into the collaboration platform through a browser.
2. The system authenticates the user's identity.
3. The user opens a document stored in cloud storage.
4. Multiple users can edit the document simultaneously.
5. Changes are transmitted through web APIs.
6. The cloud service synchronizes changes between connected users.
7. Version control records previous versions of the document.

8. Access permissions determine who can view or edit the document.

4. Security

The system provides:

- **Encryption** to protect documents during storage and transmission.
- **Identity management** to verify users.
- **Access permissions** to control document access.
- Authentication to prevent unauthorized users.
- Secure APIs for communication between clients and cloud services.

5. Benefits

- Real-time collaboration.
- Access from multiple devices.
- Automatic cloud storage.
- Document version history.
- Easy file sharing.
- High availability and scalability.

Case Study 3: Online Food Ordering Platform

1. Cloud Architecture

The Online Food Ordering Platform uses a cloud-based architecture consisting of web browsers and mobile clients, RESTful APIs, cloud backend services, cloud storage, virtual machines, and managed Kubernetes.

2. Main Components

- **Frontend Clients:** Web browsers and mobile applications.
- **RESTful Web APIs:** Connect clients with backend services.
- **Cloud Platform:** Hosts the application and backend services.
- **Cloud Storage:** Stores customer data, menu images, and invoices.
- **Virtual Machines:** Provide computing resources.
- **Managed Kubernetes:** Manages and scales application containers.

- **Security Services:** Authentication, HTTPS, and role-based access control.

3. Working

1. A customer opens the food ordering application.
2. The client sends a request to the backend using REST APIs.
3. Backend services retrieve restaurant and menu information.
4. Menu images and other information are obtained from cloud storage.
5. The customer selects food items and places an order.
6. Order information is sent to the backend through the API.
7. The backend processes and stores the order information.
8. Kubernetes automatically manages application resources according to demand.
9. The customer receives the order status through the application.

4. Security

The platform uses:

- **Authentication tokens** to verify users.
- **HTTPS** to encrypt communication.
- **Role-Based Access Control (RBAC)** to provide appropriate permissions.
- Secure cloud storage for customer information and invoices.
- API security to protect backend services.

5. Benefits

- Supports both web and mobile users.
- Scalable during high-order periods.
- Reliable cloud infrastructure.
- Secure customer transactions.
- Real-time communication.
- Automatic resource management using Kubernetes.